ASTM D913. Newly applied pavement markings shall have a minimum of ninety-seven percent (97%) of the substrate covered, determined by interpolation from the ASTM D913 Photographic References.

321.07.05 Daytime visibility – When viewed dry or wet in the daytime, the pavement markings shall be readily visible for a forward distance of 150 m, or as far forward as possible until obstructed by the road geometry if less than 150 m.

Daytime visibility will be assessed visually. Where it is not clear that the specification is met, the distance will be measured.

321.07.06 Durability – Longitudinal, Transverse, Chevron, and Crosshatch Pavement Markings shall be visually assessed for conformance with this Specification using ASTM D913.

321.07.07 Test Site and Acceptance or Rejection – A test site is defined as a site where the above six tests have been performed in a single location. If two or more line types are applied per day that pavement markings are being laid, it may be necessary to perform relevant tests for each line type at more than one location.

A failure of any one of the six above components will constitute a failure for that site and will be considered unacceptable work.

321.07.08 Quality Control – All Quality Control records shall be retained by the Contractor and made available to the Ministry Representative upon request.

The Ministry may conduct tests as deemed necessary to identify the Paint supplied is the same as the Paint submitted for qualification.

PAYMENT

321.08 General – Payment for Traffic Marking Paint and Special Markings will be made at the Traffic Marking Pavement and Special Markings contract price or per line km as itemized in the Special Provisions Schedule 7, inclusive for supply and application of Traffic Marking Paint, layout of traffic lines, glass beads, and Special Markings, costs associated with quality control, material supply, application, related material manufacturer's requirements and costs for surface preparation.

SECTION 407
FOUNDATION EXCAVATION

407.01 Scope – This Section covers foundation excavation for structures including, but not limited to, excavation for abutments, piers, walls, and bridge end fills, as specified herein, and shall apply generally to all structures other than those where it distinctly states that the prices bid are inclusive of excavation.

407.02 Excavation – All materials shall be removed as necessary for the construction of foundations or other works. Foundation excavations shall not be larger than is reasonably necessary. Excavations and adjacent highways and other facilities shall be protected as necessary by barricades, shoring, dykes and/or cofferdams.

Excavations shall be constructed in compliance with the Workers Compensation Act and the Occupational Health and Safety Regulation.

407.03 Description of Material Types – Material types will be as defined in SS 201.11.

No distinction shall be made between wet and dry excavation.

407.04 Preparation of Foundations – Care shall be taken not to disturb the bottom of the excavation. If the bottom of the excavation is disturbed in material other than rock, the Contractor shall remove and dispose of all disturbed material and shall replace it with material meeting the material, placement and compaction requirements of SS 201.40 Bridge End Fill. If the bottom of the excavation is disturbed in rock, the Contractor shall remove and dispose of all disturbed material and shall replace it with a concrete sub-footing, as directed by the Ministry Representative.

Where concrete is to be placed on rock, the rock surfaces shall be clean and free from any loose materials.

The bottom of the excavation shall be capable of providing a competent foundation and may require confirmation by the Ministry Representative prior to further works. Where, in the opinion of the Ministry Representative, the bottom of an excavation is not competent, the Ministry Representative may direct the Contractor to excavate deeper. The Ministry Representative may direct replacement of the incompetent material with material meeting the material, placement and compaction requirements of SS 201.40 Bridge End Fill or with a concrete sub-footing.

Unless underwater concreting is approved by the Ministry, excavations for concrete structures shall be dewatered, if necessary, so that concrete is placed in the dry.

407.05 Backfilling – After the structures are sufficiently built, excavations shall be backfilled to the original ground contours with excavated material, as directed by the Ministry Representative, unless this material is designated as unsuitable. In the case where backfill is to be with Bridge End Fill, the provisions of SS 202.23 shall apply.

Drainage course material shall be installed as shown on the Drawings. The gradation of the drainage course materials shall be as specified in Table 407-A.

Table 407-A: Drainage Course Material Gradation

SIEVE SIZE (mm)	% PASSING BY MASS OF TOTAL SAMPLE
40	100
20	0 – 100
10	0

407.06 Measurement and Payment – Payment for foundation excavation will be made at the Unit Price per cubic meter bid for the material types as identified in SS 407.03 and as follows.

(a) Payment shall include excavation, shoring, barricades, backfilling to the original ground contours with suitable material, compaction of the material, compaction tests and Quality Control.

(b) No payment will be made for removal and replacement of material disturbed by the Contractor below the required depth of excavation.

(c) Payment shall cover the supply and placement of gravel drainage courses, filter cloth and drain pipes, and joint waterproofing membranes if shown on the Drawings.

(d) If applicable, payment shall also cover restoration of roadway gravel and pavement damaged by the Contractor's operations.

(e) Where there is an excess of excavated materials or excavated materials are deemed unsuitable by the Ministry Representative, payment shall cover loading, hauling and disposal of such materials.

Excavated materials, which the Ministry Representative deems unsuitable for backfill, will be replaced with suitable material by the Contactor and paid for as Extra Work.

Excavation volumes for footings and box culverts shall be calculated based on the bottom of the excavations as shown on the Drawings or directed by the Ministry Representative and vertical planes 600 mm beyond the edges of the structures. No payment will be made for material removed outside of these boundaries.

No allowance shall be made for excavation of manholes or catch basins.

In the case of excavation for bridge end fill, payment will be made for material removed to the neat lines as shown on the Drawings and shall cover loading, hauling and disposal of such material.

Progress payments will be made as indicated in Table 407-B.

Table 407-B: Progress Payment Breakdown

Work	Type A	Type D
Excavation	80%	60%
Backfilling and Disposal	20%	40%

SECTION 412
CONCRETE REINFORCEMENT

DESCRIPTION

412.01 Scope – This Section describes the supply, fabrication and installation of reinforcing for concrete structures.

The requirements of CSA-S6, "Canadian Highway Bridge Design Code" and the Ministry "Bridge Standards and Procedure Manual – Supplement to CHBDC S6" shall apply unless specified otherwise herein or in the Contract.

For items not covered by CSA-S6, "Canadian Highway Bridge Design Code" and the Ministry "Bridge Standards and Procedure Manual – Supplement to CHBDC S6" or the Contract, CSA A23.1 shall apply.

References to standards shall be to the current editions at time of tendering.

MATERIALS

412.11 General – All reinforcing bars shall be deformed unless indicated otherwise on the Drawings.

The type and grade required shall be as specified on the Purchase Order, Work Order, Drawings or Special Provisions.

Unless specified otherwise, wire ties for reinforcing steel shall be minimum 1.6 mm diameter.

Two copies of mill certificates giving chemical and physical properties of the reinforcing steel shall be furnished to the Ministry Representative.

The identification of reinforcing bars shall be maintained throughout the fabrication, coating (if required) and shipping processes to the installation on the job.

The Contractor shall, as part of the Quality Control program, have random samples (selected by the Ministry Representative) of reinforcing steel, of all material types and surface treatments, tested for bending and tension. Two samples shall be taken of each material type, size and grade of bar used in the Work. Testing shall be in accordance with CSA G30.18 unless otherwise specified. Testing shall be performed by an approved testing agency.

412.11.01 Uncoated Reinforcing Steel – Concrete reinforcement shall conform to the requirements of the ASTM and CSA Standards shown in Table 412-A, as applicable.

Wire ties for uncoated reinforcing steel shall be annealed steel wire.

Table 412-A: Requirements for Uncoated Reinforcing Steel

Test Designation	Standard Test Method / Procedure
ASTM A1064	Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
CSA G30.18	Carbon Steel Bars for Concrete Reinforcement

412.11.02 Epoxy-Coated Reinforcing Steel – Epoxy-coated reinforcing steel bars shall conform to the requirements of ASTM A775 and D3963. Application plants shall be certified under the Concrete Reinforcing Steel Institute (CRSI) Epoxy Coating Plant Certification program and proof of certification shall be submitted to the Ministry Representative. All epoxy-coated bars shall be clearly labelled with the name of the manufacturer, the primer and the coating system used and the date of production. The method of labelling shall be determined by the applicator plants and test certificates of performance of epoxy coated bars shall be submitted to the Ministry Representative.

Further to paragraph 9.2 of ASTM D3963, the identification marks on all epoxy coated reinforcing bars shall be maintained from the point of delivery until they are incorporated into the structure.

The Ministry Representative may require samples of epoxy resin and epoxy patching material for acceptance testing as called for in ASTM A775. The Ministry Representative may require tests of coated bars as called for in ASTM A775.

Wire ties for epoxy coated reinforcing bars shall be epoxy coated .

412.11.03 Galvanized Reinforcing Steel – Hot dipped galvanized reinforcing bars shall conform to the requirements of ASTM A767M – Class II Coating.

Only CSA G30.18 Grade W bars shall be galvanized.

All bent bars shall be heat treated for stress relief prior to galvanizing. Stress relief shall be at a temperature from 480°C to 560°C for 1 hour per 25 mm of bar diameter as per ASTM A-767M.

Galvanized reinforcing bars shall not be placed in contact with uncoated reinforcing bars.

Wire ties for galvanized reinforcing steel shall be zinc coated or stainless steel.

412.11.04 Fibre Reinforced Polymer (FRP) – FRP bars, grids and tendons shall be manufactured and qualified in accordance with CSA S807.

The FRP supplier shall provide test data from an independent third party testing lab that demonstrates the material conforms to the specification.

Ties for FRP bar shall be plastic or nylon.

412.11.05 Low-carbon, Chromium, Steel Reinforcing Bar – Low carbon, chromium reinforcing bar and spirals shall conform to ASTM A1035, Type CS, grade 100 (690 MPa). Nominal dimensions, unit masses and deformation requirements for Imperial bar sizes shall conform to ASTM A1035.

Unless otherwise specified, only one type or grade of low-carbon, chromium, reinforcing steel shall be supplied for use throughout the project.

Wire ties for low carbon, chromium reinforcing bar shall meet the requirements for tie wire for stainless steel reinforcing bar.

412.11.06 Stainless Steel Reinforcing Bar– Stainless steel reinforcing bars and spirals shall conform to ASTM A276 and ASTM A955, minimum Grade 420, except as noted. Nominal dimensions, unit masses and deformation requirements for Imperial bar sizes shall conform to ASTM A955.

Stainless steel reinforcing bars shall be of a stainless steel type specified in Table 412-B.

Unless otherwise specified, only one type of stainless reinforcing steel shall be supplied for use throughout the Work.

Stainless steel reinforcing bars shall be hot-rolled, de-scaled and pickled to the required mechanical properties and dimensions.

Stainless steel reinforcing bars shall be free of deposits of iron and non-stainless steel.

Fabrication of stainless steel reinforcing bars shall be such that the bar surfaces are not contaminated with deposits of iron and non stainless steel and that the surfaces are not damaged due to straightening from coil.

Wire ties for stainless steel reinforcing bars shall be stainless steel wire of a type listed in Table 412-B except that tie wire used to tie stainless steel reinforcing bars to epoxy coated reinforcing steel bars shall be epoxy coated wire.

412.11.07 Order Lists and Bending Diagrams – All order lists and bending diagrams shall be submitted to the Ministry Representative for review before material is ordered.

The review of order lists and bending diagrams by the Ministry Representative shall not relieve the Contractor or Suppliers, as applicable, of responsibility for the correctness thereof.

Table 412-B: Types of Stainless Steel

Type	ASTM Designation	Unified Numbering System (UNS) Designation
316 LN	A955	S31653
DUPLEX 2205	A955	S31803
DUPLEX 2304	*	S32304

** Chemical composition shall be within the ranges identified in Table 1 (Typical Composition %) of and in conformance to the requirements of ASTM A276. The dimensioning and mechanical properties shall be in conformance with ASTM A955.*

CONSTRUCTION

412.31 Bending – Reinforcement bars shall be cut and bent to the shapes shown on the Drawings. Bending shall be sufficiently accurate that the placing tolerances (specified in SS 412.33.02) can be met. All bars shall be bent cold, unless otherwise permitted. Bars partially embedded in concrete shall not be field bent except as shown on the Drawings or where specifically permitted.

Field bending of galvanized steel reinforcing bar and fibre reinforced polymer reinforcing bar is not allowed.

Field bending of stainless steel reinforcing bar is only allowed upon prior written approval from the Ministry's Representative. Stainless steel reinforcement shall only be bent using equipment specifically designed for that purpose.

412.31.01 Hooks and Bend Dimensions – Unless otherwise shown on the Drawings, hooks shall have the following dimensions:

(a) 180° bend plus extension of at least 4 bar diameters, but not less than 60 mm.

(b) 90° bend plus extension of at least 12 bar diameters.

(c) for stirrups and ties only, either a 90° or a 135° bend plus extension of at least 6 bar diameters at the free end of the bar.

(d) Minimum inside diameters of bends shall be as shown in Table 412-C. For stainless steel reinforcing bars #8 and smaller, the minimum bend diameters shall be six times the bar diameter. Minimum bend diameters for #9 through #12 stainless steel reinforcing shall be eight times the bar diameter. Minimum bend diameters for #13 and larger stainless steel reinforcing shall be ten times the bar diameter.

(e) Exceptions:

(i) Minimum inside diameters of bends and 90º and 135º hooks for stirrups and ties shall be 4 bar diameters for uncoated and stainless steel bars and 8 bar diameters for epoxy-coated bars.

(ii) Minimum inside diameters of bends in welded wire fabric, plain or deformed, for stirrups and ties shall not be less than 4 wire diameters for deformed wire larger than 7 mm and two wire diameters for all other wires, except that bends with an inside diameter of less than 8 wire diameters shall be not less than 4 wire diameters from the nearest welded intersection.

Note: Galvanized reinforcing bars shall be pre-bent before galvanizing.

Table 412-C: Minimum Inside Diameters of Bends

Bar Size	Uncoated (mm)			Epoxy coated, FRP, and Galvanized (mm)	Stainless Steel* (mm)
	300R	400R or 500R	400W or 500W		
10M	60	70	60	80	60
15M	90	100	90	120	90
20M	-	120	100	160	120
25M	-	150	150	200	150
30M	-	250	200	240	240
35M	-	300	250	350	280
45M	-	450	400	450	450
55M	-	600	550	550	550

**for imperial stainless steel bar sizes see text in SS 412.31(e)*

412.32 Handling and Storage – Reinforcing bars shall be stored on platforms, skids or other suitable supports clear of the ground and shall be protected as practicable from mechanical injury and surface deterioration caused by exposure to conditions producing rust. When placed in the Work, reinforcement shall be free from dirt, loose rust or scale, mortar, paint, grease, oil, or other materials that would reduce bond.

412.32.01 Epoxy-Coated Reinforcing Steel – Epoxy-coated reinforcing bars shall be handled and stored so as to minimize damage to the coating. Equipment for handling the bars shall have protected contact areas. Nylon slings or padded wire rope slings shall be used. Suitable bundling bands shall be used to prevent damage. Bundles of coated bars shall be lifted at multiple pick-up points to prevent bar-to-bar abrasion from sags in the bundles of bars. The bars or bundles shall not be dropped or dragged.

Epoxy coated steel reinforcing bars shall be stored off the ground on protective cribbing, and timbers placed between bundles when stacking is necessary. Supports shall be placed sufficiently close to prevent sags in the bundles.

Long term storage of epoxy-coated bars shall be minimized.

If circumstances require storing epoxy-coated steel reinforcement outdoors for more than two months, protective storage measures shall be implemented to protect the material from sunlight, salt spray and weather exposure. Epoxy-coated steel reinforcing bars stored in corrosive environments will require protection sooner. Epoxy-coated steel reinforcing bars or bundles shall be covered with opaque polyethylene sheeting or other suitable opaque protective material.

Epoxy-coated bars shall not be cut or bent on Site without the permission of the Ministry Representative. In any case, the flame-cutting of coated bars will not be permitted.

When immersion-type vibrators are used to consolidate concrete around epoxy-coated steel reinforcing bars, the vibrators shall be equipped with rubber or non-metallic vibrator heads.

Any bar with coating damage exceeding 2% of the surface area of the coated steel reinforcing bar in any 300 mm length, shall be rejected. When the extent of the damage does not exceed 2% of the surface area in any 300 mm length, all damaged coating shall be repaired with patching material in accordance with ASTM A775, Section 12. Patching material shall be applied in strict accordance with the manufacturer's written instructions. Prior to application of the patching material, rust shall be removed from the damaged areas. The patching material shall be allowed to cure before placing concrete over the epoxy-coated steel reinforcing bars.

Compliance with the "Guidelines for Jobsite Practices", contained as an Appendix of ASTM A775 shall be mandatory.

Where on-site inspection or testing by the Ministry Representative demonstrates non-compliant coating thickness or integrity, for any shipment of bars, such bars may be rejected, and in any case shall not be placed in the structure until the Ministry Representative is satisfied as to their acceptability.

The Contractor shall make available to the Ministry Representative, for inspection and acceptance, all repairs performed to protective coatings on any reinforcing steel. Repairs of epoxy coatings shall not be performed when the epoxy-coated bar or ambient air temperature is 5°C or less, or when moisture is present on the bar. Patching material shall not be applied if precipitation is expected within four hours of the procedure. If precipitation does occur within four hours, the Ministry Representative shall determine as to whether the material as applied is acceptable. If the

placed epoxy-coated bars have been exposed to salt spray or road salts, the bars shall be rinsed with fresh water to remove any chloride contamination prior to placing concrete.

412.32.02 Stainless Steel Reinforcing Bar – Stainless steel reinforcing bars shall be stored clear of the ground on timbers or other suitable protective cribbing spaced to prevent sags in the bundles. Stacks of bundles of straight bars shall have adequate blocking to prevent contact between the layers of bundles. Stainless steel reinforcing bars shall be stored separately from black reinforcing steel bars, with the bar tags maintained and clearly visible until ready for placing.

412.33 Placing and Fastening – Before any concrete is placed, the placing and securing of reinforcing steel including dowels, within the area of concrete placement shall be complete. Tying in place of all dowels projecting from the area of concrete placement shall be subject to acceptance by the Ministry Representative before any concrete is placed. The reinforcing steel shall be free from dirt, detrimental rust, loose scale, paint, oil or other foreign material. Steel reinforcement - other than stainless steel reinforcing bars - with rust, mill scale, or a combination of both shall be acceptable, provided the minimum physical properties including height of deformations and mass of a wire brushed test specimen are not less than the applicable specification requirements.

Reinforcement shall be placed in the positions shown on the Drawings, within the tolerances specified below, adequately supported and secured against displacement. All splices of adjacent bars shall be securely tied together. Tying 100% of the bar intersections is required where the bar spacing centres are 300 mm or greater. Tying of 50% of the bar intersections is required when the bar spacing centres are less than 300 mm.

The locations of the top reinforcing steel in bridge decks shall be checked by running a full deck-width template along the longitudinal screeds. The lower edge of the template shall be set at the cover dimension (as specified or noted in the Drawings) plus any anticipated deflection of the screed rail support system below the level of the deck surface. No steel shall touch the template nor be more than 6 mm distant from it. Any discrepancies outside of these limits shall be corrected by the Contractor to the satisfaction of the Ministry Representative.

The location of reinforcing steel near deck joint anchors shall be adjusted so that there will be no interference with the deck joint anchors.

412.33.01 Tack Welding – Tack welding of reinforcement for cage assembly or securing of reinforcement will be permitted only in the following circumstances:

(a) on bars provided by the Contractor that are additional to design requirements, except that no additional shear stirrups or tensile reinforcement bars will be permitted, as they may adversely affect structural performance.

(b) at the ends of bars where the Contractor provides additional length that is in excess of design requirements and this additional length is detailed on bar bend sheets or shop drawings, with notes indicating the purpose of the additional length. Such bar bend sheets or shop drawings shall have written acceptance by the Ministry Representative prior to welding taking place.

(c) with the written acceptance of the Ministry Representative, in which case, welding shall be in accordance with SS 412.34.02.

412.33.02 Tolerances – Tolerances for placing reinforcement shall, unless otherwise specified, be as shown in Table 412-D.

Table 412-D: Tolerances for Placing Reinforcement

Element	Tolerance
Concrete cover to top reinforcing steel in bridge decks	+ 6 mm – 0 mm
Concrete cover to reinforcement, other than top reinforcing steel in bridge decks	± 8 mm
Bar location, except cover, when depth of a flexural member, thickness of a wall or smallest dimension of a column is 200 mm or less	± 8 mm
Bar location, except cover, when depth of a flexural member, thickness of a wall or smallest dimension of a column is larger than 200 mm but less than 600 mm	± 12 mm
Bar location, except cover, when depth of a flexural member, thickness of a wall or smallest dimension of a column is 600 mm or larger	± 20 mm
Longitudinal location, except cover, of bends and ends of bars	± 50 mm
Longitudinal location, except cover, of bends and ends of bars at discontinuous ends of members	± 20 mm

412.33.03 Bar Supports and Spacers – Bar supports and spacers shall be adequate to ensure concrete cover and bar spacings are maintained within the specified tolerances.

Bar supports and spacers shall be sufficient in number and strength to support the reinforcement and prevent displacement by workers or equipment before and during concreting and shall be adequately spaced to ensure that any

sagging between supports does not intrude on the specified concrete cover.

Bar supports and spacers shall be of a type and material that will not cause rust spots, blemishes or spalling of concrete surfaces.

Bar supports and spacers shall be precast concrete, plastic, hot-dip galvanized steel, epoxy-coated steel or stainless steel, except that, plastic supports or plastic spacers over 200 mm in height shall not be allowed.

Precast concrete supports shall have a compressive strength and quality not less than that of the concrete in which they are embedded. For finished surfaces, the face of the support in contact with the forms shall not exceed 50 mm in any dimension and shall have a colour and texture to match that of the finished concrete surface.

Bar supports and spacers for approach slabs and formed horizontal slabs such as bridge decks, bridge sidewalks, and top slabs of culverts shall meet the following additional requirements:

(a) Supports or spacers up to 200 mm in height shall be either precast concrete or plastic.

(b) Plastic supports or spacers shall have an individual minimum breaking strength of 4.5 kN (1000 lbs) and shall be capable of maintaining strength and dimensional properties for the range of temperatures encountered on Site.

(c) Maximum spacing between supports, for each mat of bars, shall be 1200 mm, except that the maximum spacing between plastic supports, for each mat of bars, shall be 1000 mm.

(d) Each support shall carry the load from not more than one mat of reinforcing steel, except as provided below:

(i) Where a support is used that is specifically designed by the manufacturer to carry two mats at two separate positions, or

(ii) Where the top mat of reinforcing steel is more than 200 mm above the slab soffit, the top mat may be supported by reinforcing steel spreader bars, matching the material of the top mat. Spreader bars shall be secured between the top and bottom reinforcing mats and the bottom reinforcing mat shall be supported on precast concrete chairs adequately spaced to carry the additional weight of the top mat of reinforcing steel.

412.33.04 Concrete Cover – Concrete cover for reinforcing steel shall be in accordance with the Ministry "Bridge Standards and Procedure Manual – Supplement to CHBDC S6".

412.33.05 Spirals – Spirals shall be held in place and to line by vertical spacers.

(a) For spiral rods less than 15M, spacing shall be as follows:

(i) two spacers per loop for spirals less than 500 mm in diameter.

(ii) three spacers per loop for spirals 500 mm to 800 mm in diameter.

(iii) four spacers per loop for spirals over 800 mm in diameter.

(b) For spiral rods 15M and larger, spacing shall be as follows:

(i) three spacers per loop for spirals up to 600 mm in diameter.

(ii) four spacers per loop for spirals over 600 mm in diameter.

(c) If vertical reinforcements are to serve as spacers, spirals shall be securely tied to vertical reinforcement at the spacing required for spacers.

(d) Stainless spirals shall be provided with stainless steel spacers in conformance with ASTM A276.

412.34 Splicing of Bars – Special requirements for splicing, such as particular locations for splices, use of overlength bars or special lap lengths, shall be as shown on the Drawings.

Splices in bars larger than 35M shall be mechanical coupler splices or welded. The detail of such splices shall be subject to acceptance by the Ministry Representative.

412.34.01 Lapped Splices

(a) Horizontal bars shall not be spliced unless greater than standard mill lengths – 12 m for 10M bars and 18 m for larger bars – would otherwise be required.

(b) Splices to be staggered so that no more than one third of the reinforcing steel in a member shall be spliced in any transverse section that is within the required lap length.

(c) If not shown on the Drawings, the length of lapped splices shall be in accordance with CSA-S6, Clause 8.15.10.1.

(d) Adjacent reinforcing bars shall not be spliced at the same locations unless shown on the Drawings or authorized by the Ministry Representative.

(e) Column spirals shall be spliced in accordance with CSA-S6, Clause 8.14.4.2.

(f) Lap splice bars shall be of the same material as the bars being connected.

412.34.02 Welded Splices – Welded splices shall be used only if detailed on the Drawings. Welding shall conform to the requirements of CSA W186, and shall be performed by a company certified by the Canadian Welding Bureau to the requirements of CSA Standard W186.

Welded splices shall not be used on epoxy coated bars or stainless steel bars, with the exception of stainless steel spirals.

412.34.03 Mechanical Coupler Splices – Mechanical couplers shall be used for splices only if pre-approved or detailed on the Drawings or authorized in writing by the Ministry Representative.

(a) **Coupler Yield Strength –** Such couplers shall develop in tension or compression, as required, at least 120% of the specified yield strength of the bars, but not less than 110% of the mean yield strength, representative of the bars to be used, in the test of the mechanical connection.

(b) **Wedge Couplers–** Wedge couplers shall not be used in bars greater than 15 mm diameter.

(c) **Concrete Cover –** When mechanical couplers are used, the Contractor shall ensure that the minimum concrete cover as specified is maintained.

(d) **Stainless Steel Couplers** – Where mechanical couplers are to be used to couple stainless steel reinforcing bars, the mechanical coupler shall be stainless steel of a type specified in Table 412-B.

(e) **Allowable Slip –** The total slip of the reinforcing bars within the splice sleeve of the connector after loading in tension to $0.5f_y$ and relaxing it to $0.05f_y$ shall not exceed the following measured displacements between gauge points straddling the splice sleeve:

(i) Up to and including 45M bars 0.25 mm

(ii) 55M bars 0.75 mm

MEASUREMENT

412.81 General – When required, reinforcing steel will be measured by KILOGRAMS for the total mass in each part of the structure(s) as computed using Table 412-E. The mass of clips, separators, tie wires, chairs and other fastening devices, and epoxy or other coating shall not be added into the calculations of mass of reinforcing steel.

PAYMENT

412.91 General – Payment for REINFORCING STEEL will be made at the Contract Lump Sum Prices for reinforcing steel in each part of the structure(s), for the supply, cutting, bending, coating (if required), shipping, placing and securing of reinforcing steel. Payment shall also include all costs associated with Quality Control.

412.92 Payment for Changed Quantities Ordered by the Ministry – Payment for increases or reductions in quantities of uncoated and coated reinforcing steel ordered by the Ministry Representative will be made as a Change to Work in accordance with GC38.00.

Table 412-E: Nominal Linear Mass for Reinforcing Steel

Bar Designation	Mass (kg/m)
10M	0.785
15M	1.570
20M	2.355
25M	3.925
30M	5.495
35M	7.850
45M	11.775
55M	19.625
#3 (stainless steel)	0.556
#4 (stainless steel)	1.011
#5 (stainless steel)	1.559
#6 (stainless steel)	2.225
#7 (stainless steel)	3.032
#8 (stainless steel)	3.995
#9 (stainless steel)	5.053
#10 (stainless steel)	6.416
#11 (stainless steel)	7.880

SECTION 413
BRIDGE DECKS AND CONCRETE OVERLAYS

DESCRIPTION

413.01 General – This Section describes requirements for the construction of Bridge Decks, and for the construction of Concrete Overlays on previously constructed (old) bridge decks and on newly constructed (new) bridge decks. The requirements of SS 211 shall apply except as otherwise specified in this Section. Unless otherwise specified in this Section, bridge deck concrete shall include diaphragms, deck slabs, bridge parapets and medians, bridge sidewalks, approach slabs and overlay concrete.

413.02 Bridge Decks

413.02.01 General – Deck slab construction shall be full-width and continuous between the joints as shown on the Drawings, and shall be placed in the sequence as shown on the Drawings. Any variation from this requirement shall be subject to acceptance by the Ministry Representative.

(a) Camber Survey – Prior to commencing deck construction and subsequent to the completion of the erection of girders, with the girders in a thermally neutral condition and with no dead load other than the dead load of the girders and necessary fall protection, the Contractor shall profile all the girders. Camber elevations are to be taken at 1000 mm centres maximum (or as indicated on the Drawings or Special Provisions), centrelines of all bearings and ends of girders. The camber elevation locations are to be laid out to an accuracy of 25 mm horizontally and camber elevations to 1 mm vertically. Within 14 days of the Contractor supplying the camber elevations to the Ministry Representative, the Contractor will receive haunch heights for setting the deck slab soffit, (for suspended slab type structures only) and the design finished grade elevations for the deck, complete with anticipated dead load deflections at 1000 mm stations along the centreline of roadway, to be used by the Contractor in establishing the required screed rail settings.

In the event that actual girder camber values vary significantly from the design values, the Contractor will be required to modify the original design finished grade as directed by the Ministry Representative.

(b) Method, Equipment & Personnel – The Contractor shall submit to the Ministry Representative for review, four weeks prior to the scheduled start of deck slab concrete placement, a Deck Placement Plan which provides a complete description of the method, equipment, material and personnel, including previous deck placement experience of the personnel to be employed. This shall include the equipment for mixing, transporting, placing, vibrating and curing the concrete and shall also include details of the finishing equipment and support rails. The submittal shall clearly demonstrate that the proposed methods, equipment and personnel have the capability to carry out the Work in accordance with SS 413. All review concerns raised shall be addressed to the satisfaction of the Ministry Representative. All equipment shall be subject to inspection and acceptance by the Ministry Representative. Equipment may be rejected during operation if it is deemed unsatisfactory.

Two weeks prior to the scheduled start of deck slab concrete placement, the Contractor shall meet with the concrete supplier and the Ministry Representative to review deck concrete operations with respect to concrete mix design and performance, concrete delivery, placing, finishing and curing equipment and methods.

(c) Haunch Setting, Screed Rails & Reinforcing Cover – The Contractor shall set the deck slab soffit formwork to within 2 mm of the haunch heights as determined in accordance with SS 413.02.01(a).

After installation of the deck slab reinforcing, the Contractor shall set the screed rails for the deck finishing machine. Screed rails shall be installed to an accuracy of ±2 mm and have a maximum deflection between supports of 1 mm. Allowance for dead load deflection of the structure shall be made in the screed rail setting as well as any anticipated deflection due to the loading of any formwork that may be supporting the screed rail system. The screed rails shall be adequately supported outside the area to be concreted and shall be completely installed before concrete is placed, unless otherwise accepted by the Ministry Representative.

Prior to the placement of deck slab concrete and subsequent to the installation of deck slab reinforcing and screed rails, the locations of the top reinforcing steel in bridge decks shall be checked by running a full width deck template along the longitudinal screeds. The lower edge of the template shall be set at the cover dimension (as specified or noted in the Drawings) plus any anticipated deflection of the screed rail support system below the level of the deck surface. No steel shall touch the template nor be more than 6 mm distant from it. Any discrepancies outside of these limits shall be corrected by the Contractor to the satisfaction of the Ministry Representative.

413.02.02 Concrete Placement

(a) Temperature Control and Thermal Control at Placement – Concrete designed for bridge decks and overlay including any bridge deck concrete containing silica fume shall not have a temperature in excess of 20°C at time of placement.

Mass concrete shall meet the requirements of CSA A23.1. Mass concrete shall apply to concrete placements having a minimum dimension equal to or greater than 1 m. The Contractor shall submit a thermal control plan to the Ministry Representative for review and acceptance at least 2 weeks prior to concrete placement for mass concrete. The thermal control plan for mass concrete shall be in accordance with CSA A23.1.

Methods used for maintaining temperature controls shall not diminish the quality of the concrete, and shall not exceed the specified w/cm ratio.

When the atmospheric temperature is 20°C or higher, SS 211.18 – Hot Weather Concreting, shall apply except as modified below:

(i) Concrete for bridge decks and overlays shall not have a temperature greater than 17°C when placed. Prior to the placement of bridge deck and overlay concrete, when the ambient air temperature is at or above 20°C, or when there is a probability of its rising to 20°C or above during the placement period (as forecast by the nearest official meteorological office), the Contractor shall submit to the Ministry Representative for acceptance, the proposed placing operations complying with the requirements for hot weather concreting.

(ii) When the ambient air temperature is at or above 20°C, or when there is a probability of its rising to 20°C or above during the placement period (as forecast by the nearest official meteorological office), bridge decks and overlays shall be cast when daily air temperatures are forecast to be a minimum (normally between 12:00 midnight and 9:00 AM).

(b) Time of Placement – All bridge deck and overlay concrete shall be fully discharged and placed in its final location within 90 minutes after the water and cement have been combined. Under conditions contributing to rapid stiffening of concrete, the Contractor shall adjust its placing operations to suit reduced handling time.

Time of placement extensions will be considered on a case-by-case basis and may require the use of time extension admixtures such as Hydration Stabilizing Admixtures (HSAs) conforming to the requirements of ASTM C494 Type B, retarding or Type D, water-reducing and retarding, and retarding admixtures.

When accepted by the Ministry Representative, the use of set retarders or HSA's shall be as follows:

(i) Concrete for bridge decks shall be fully discharged and placed within 180 minutes after water and cement have been combined.

(ii) Time extensions are subject to preconstruction trials being conducted by the Contractor, to establish the appropriate admixture dosage to provide suitable extended slump life and set time of concrete, without increasing the water/cementitious ratio of the concrete above that which would be required if time extension admixtures were not used. Slump retention testing shall be carried out to verify the extended slump life. Set time testing shall be carried out in accordance with ASTM C403. The use of time extension admixtures shall in no instance modify the maximum concrete temperature required at time of placement.

(iii) A one-time only addition of time extension admixture will be allowed. This will be during initial batching of the concrete or immediately at completion of batching as recommended by the admixture manufacturer. Addition of time extension admixture at any other time will be cause for rejection of the concrete.

(c) Concrete Placement - Diaphragms – Unless otherwise specified, diaphragms shall be placed either:

(i) not more than two hours before the adjacent deck concrete is placed. In this case, diaphragm concrete shall have a slump less than 40 mm and shall be retarded so that it sets at the same time as the deck concrete, or

(ii) a sufficient length of time ahead of the deck concrete, so that the diaphragm concrete attains a compressive strength of 15 MPa before the deck slab is cast.

Sequence for the placing of concrete for diaphragms at ends of girders on continuous and integral type spans shall be as shown on the Drawings.

(d) Concrete Placement - Deck Slab – The top surfaces of concrete stringers and deck panels, if applicable, shall be prepared in accordance with SS 211.12.03 and the surfaces shall be in a saturated surface dry condition immediately before the deck is cast.

Concrete shall be placed at a rate of not less than 6 m/hr. The concrete front and finishing operation shall be kept parallel to the substructure. Placing shall be started at such time as to permit finishing during daylight hours, except if ambient temperatures are as outlined in SS 413.02.02(a), then adequate lighting is

to be provided in accordance with Workers Compensation Act and the Occupational Health and Safety Regulation, to allow for concrete operations under reduced light conditions.

In the event of unexpected rain, concrete placing shall cease and the surface shall be protected immediately. Surface concrete which has been rained on shall immediately be removed to the top of reinforcing steel.

(e) Finishing – The deck surface shall be finished by a drum-type deck finishing machine on support rails. Acceptable finishing machines are Terex Bid-Well models 2450, 3600 or 4800, Gomaco models C-450, C-750, Allen Models 4836B, 6036 B 6048 B or alternate acceptable to the Ministry Representative.

There shall be no disturbance of the concrete after it has been consolidated and finished by the deck finishing machine. Two work bridges shall be provided to carry out the final deck finishing operations and for preparations to cure the concrete deck surface. No inserts will be permitted in the finished concrete roadway surface.

The surface behind the finisher shall be floated transversely to eliminate ridges and open texture. Areas, which cannot be machine-finished, shall be hand screeded and floated.

Prior to final surface finishing or texturing, the deck surfaces shall be repeatedly checked for accuracy of deck slab thickness and alignment. The Contractor shall measure and record the deck slab thickness at a frequency of at least one measurement per interior bay and overhang bay, at a maximum of 3000 mm centers longitudinally. Adjustments as acceptable to the Ministry Representative shall be made to the deck-finishing machine to maintain the minimum deck slab thickness and the required concrete cover on the deck reinforcing. The thickness readings of the deck slabs shall be supplied to the Ministry Representative at the conclusion of each placement section.

For bridge decks that will not have a waterproofing membrane, the final surface texturing shall be tined. The tining shall create transverse grooves 3 mm wide by 1.5 mm to 3 mm deep at 20 mm centre-to-centre spacing.

For bridge decks that will have a waterproofing membrane, the final concrete surface shall be floated and troweled as necessary to provide a smooth surface.

413.02.03 Repairs – Areas of the deck not meeting the tolerances as given in SS 211.16, shall be corrected as follows:

(a) where the required alignment can be obtained with a maximum cut of 6 mm, the bumps shall be cut with a bump cutter which will provide an acceptable longitudinal grooved texture;

(b) where the required alignment cannot be obtained with a maximum cut of 6 mm, the concrete shall be broken out with vertical edges to a minimum depth of 40 mm and shall be replaced with new concrete of the required quality.

Any physical damage to the deck surface shall be cut out and replaced. Any areas of open texture or plucked aggregates shall be repaired to the satisfaction of the Ministry Representative.

413.02.04 Interim Strength Requirements – Unless otherwise approved by the Ministry Representative, in consultation with the Professional of Record for the structure:

(a) parapet pours shall not commence until

(i) all deck pours are complete, and

(ii) the deck concrete strength exceeds 15 MPa; and

(b) heavy loads, such as concrete trucks, shall not be allowed on the structure until the deck strength exceeds 25 MPa.

413.02.05 Curing

(a) General – Freshly deposited concrete shall be protected from exposure to freezing, cold temperatures, abnormally high temperatures or temperature differentials, premature drying, excessive moisture, moisture loss, heavy shocks, excessive vibrations and high stresses, for the period of time necessary to develop the required concrete properties.

(i) The Contractor shall submit a detailed Deck Curing Plan for curing the deck and overlay concrete. The plan shall contain details such as water source, the distribution system, the collection and run-off control system, person(s) responsible for quality control, and staff to maintain the system. The curing system shall include burlap and be capable of maintaining a free film of water on the areas of concrete requiring curing, for a minimum of 8 unattended hours and this capability shall be demonstrated prior to concrete placement. It shall be the responsibility of the Contractor to ensure that the system of curing and protection is properly planned, constructed and maintained throughout the entire curing period.

(ii) Burlap shall be pre-soaked by immersing it in water for a period of at least 24 hours immediately prior to placing.

(iii) Two layers of burlap shall be applied immediately after finishing of the concrete surface within 2 to 4 m of the final surface texturing or finishing operations (no more than 20 minutes) and shall be maintained in a saturated condition in a manner, which does not

damage the finished concrete surface. Strips must overlap 150 mm and must be held in place without marring the surface of the concrete. The burlap shall be maintained in a continuously wet condition throughout the curing period and a continuous film of free water on the surfaces of the areas noted herein shall be maintained by means of soaker hoses, sprinklers, etc.

(iv) Concrete for diaphragms, monolithic decks, concrete overlays, approach slabs, parapets, and sidewalks shall be cured for a minimum of seven consecutive uninterrupted twenty-four hour periods at a minimum temperature of 10°C and for the time necessary to attain 80% of the specified compressive strength of the concrete. The compressive strength tests for verifying completion of the curing period shall be determined using field cured test cylinders cured in accordance with Clause 9.3.2 of CSA A23.2-3C. The curing period will begin following the completed placement of the concrete in the section cast.

(v) The temperature of the concrete being cured must be monitored to ensure maximum and minimum temperature requirements in accordance with SS 211.15.02 are met.

(vi) Regardless of ambient temperature, moist curing with burlap and water must be provided at all times. During cold weather, burlap must be prevented from freezing.

(vii) At completion of the curing period, the curing shall be discontinued in a manner so as not to subject the concrete to undue stresses due to rapid moisture loss in the concrete.

(viii) When the air temperature is at or below 5°C or when there is a probability of it falling below 5°C within the next 5 days after the curing period, then the following shall apply:

(A) At completion of the curing period, the addition of free water is to be discontinued and followed by an additional five (5) consecutive uninterrupted twenty-four hour periods of air drying with the concrete temperature maintained at a minimum temperature of 5°C.

(b) Additional Curing Requirements for Concrete Containing Silica Fume, Fly Ash or Slag, With or Without Fibres – A fog mist shall be applied continuously from the time of screeding until the concrete is covered with burlap, in such a way as to maintain high relative humidity above the concrete and prevent drying of the concrete surface. Water must not be allowed to drip, flow, or puddle on the concrete surface during fog misting, when placing the burlap, or at any time before the concrete has achieved final set.

Fogging equipment will consist of pressure washers (minimum output pressure of 28 MPa (4000 psi)) with suitable tips for producing a fine water mist or fog. A minimum of two pressure washers shall be used for this process and the adequacy of the range and coverage shall be demonstrated prior to placing any concrete.

When the air temperature is at or below 5°C or when there is a probability of it falling below 5°C within the next ten (10) days after the curing period, then the following shall apply:

(i) At completion of the curing period, the addition of free water is to be discontinued and followed by additional ten (10) consecutive uninterrupted twenty-four hour periods of air drying with the concrete temperature maintained at a minimum temperature of 10°C.

Modifications to the curing process may be required if it is anticipated that the deck may be subjected to the application of de-icing chemicals within the first month after the curing period.

413.02.06 Payment – for bridge deck FORMWORK and CONCRETE will be made as outlined in SS 211.21 01 and SS 211.21.02 respectively and payment for bridge deck REINFORCING STEEL will be made as outlined in SS 412.91 unless otherwise provided for in SS 413.

413.03 Concrete Overlays

413.03.01 413.32.01 General – This Subsection describes requirements for repair and resurfacing of existing bridge decks (old) and newly constructed bridge decks (new) with a concrete overlay, describing the construction practices, materials and equipment required to place acceptable concrete overlay. This work is usually carried out on one-half of the deck at a time, while traffic is maintained on the other half. The requirements of SS 413.02 shall apply except as otherwise specified in this Subsection.

413.03.02 Gradeline – The Contractor shall survey and reference profiles of the existing bridge deck and approaches prior to scarification as follows.

(a) Profile Surveys

(i) Longitudinally the profiles shall be taken at 2 to 3 m stations for the entire length of the structure, including 21 m beyond each end, with additional shots taken at the back of ballast walls and all existing or proposed new joint locations.

(ii) Transversely the number of profiles required will be determined by:

(A) A maximum spacing between elevation points of 3 m.

(B) Profiles are required 500 mm from each curb or parapet face.

(C) Profiles are required 300 mm (inside area to be concreted) from longitudinal overlay construction joint lines requiring adjustable screed rails. In general longitudinal overlay construction joints shall be located at the center of a lane or between lanes. The final location of the longitudinal construction joints is subject to the acceptance of the Ministry Representative.

(D) A profile is required at hinge point of crown.

(iii) Elevation points are to be laid out in cross section ninety degrees to centerline of roadway on straight structures and radial to centerline of roadway on curved or spiraled structures.

(iv) Profile elevation points shall be laid out to an accuracy of ±25 mm horizontally and elevations taken to an accuracy of ±3 mm vertically and be referenced such that they can be re-established within ±25 mm from their original location.

(b) The Contractor shall provide the Ministry Representative with the gradeline profiles plotted at a scale of 1:10 vertical and 1:50 horizontal and a spreadsheet showing actual original elevations for all points surveyed.

(c) In the case of old decks, subsequent to scarifying and cleaning of the deck surface, the Contractor shall re-establish the original elevation points, take new elevations for the points to the nearest millimeter and submit these elevations along with plotted profiles to the Ministry Representative.

(d) The Ministry will provide the Contractor with design finished grade profiles and elevations for the survey points as originally laid out by the Contractor. These profiles and elevations shall be used by the Contractor to set the screed rails for the deck-finishing machine, checking of overlay thickness during the dry run of the deck machine and for calculating the volume of overlay concrete required.

(e) The Contractor is responsible for properly setting the screed rails to match the design gradeline provided by the Ministry Representative. Depressions creating ponded water, or localized high spots in the concrete surface, resulting from deficient finishing procedures shall be repaired at the Contractor's expense.

(f) At completion of the deck overlay, the Contractor shall re-establish the original elevation points, take as built elevations for the points to the nearest millimetre, and submit these elevations along with plotted profiles to the Ministry Representative.

(g) No specific payment shall be made for this work as all costs associated with the production of the gradelines and profiles are deemed incidental.

413.03.03 Removal of Existing Concrete Deck Surface – The existing concrete surface shall be removed by use of scarification (roto-milling), hydro-demolition or alternate methods acceptable to the Ministry Representative, over the full length and width of the bridge deck. If methods other than scarification are proposed by the Contractor, the Contractor shall submit all required details of the proposed work to the Ministry Representative and shall obtain the acceptance of the Ministry Representative at least two weeks prior to starting the work. Alternate methods shall meet the requirements specified for scarification.

(a) **Scarification –** If scarified, equipment shall be capable of concrete removal to an accuracy of ±5 mm. The scarifying equipment shall produce a surface that does not exceed 10 mm in amplitude for roughness. The maximum cut will be to within 10 mm of the reinforcing steel. Depth of scarification will be specified by the Ministry.

The Contractor shall monitor the total depth of concrete cover to the top mat of reinforcing steel using a pachometer or alternate methods acceptable to the Ministry Representative. Readings shall be taken at regular intervals ahead of the scarification. The Contractor shall adjust the rate and depth of concrete removal to ensure that the reinforcing steel is not damaged and the desired scarified profile is achieved. All reinforcing steel damaged by the Contractor's operations will be replaced at the Contractor's expense. Care must be taken not to damage the existing deck drains. The Contractor will be required to repair or replace any damaged drains at the Contractor's expense. Drains are to be plugged to insure containment of debris.

All areas inaccessible to the scarification equipment shall be prepared by hydro-demolition, grinders or by jackhammers with weight limited to a maximum 14 kg. Preparation shall minimize damage to the sound substrate concrete. Larger jackhammers may be used if accepted by the Ministry Representative and if performance acceptable to the Ministry Representative can be demonstrated.

In the case of previously overlaid decks, concrete shall be removed to below the old bond line. Where the bond line is within 10 mm of the reinforcing steel, the remaining concrete and the old bonding material shall be removed by hydro-demolition or pneumatic tools.

Upon completion of scarification of the deck, prior to opening the scarified deck lane to traffic and at the direction of the Ministry Representative, the Contractor shall place temporary asphalt tapers (minimum 40:1) using asphalt patch material (cold mix) with tar paper underneath at all deck joints and

bridge ends as required to maintain adequate traffic movement. The asphalt tapers shall be maintained until traffic is no longer using that lane or the final paving is complete. The temporary tapers shall be in accordance with SS 502.

(b) Payment – Payment for removal of the existing concrete deck surface and asphalt roadway approaches will be made at the Lump Sum price bid. Payment for removal shall include all work for the removal of the concrete and asphalt roadway approaches including but not limited to sweeping, washing and /or blowing of debris, power hand tool removal of all concrete inaccessible to the scarification process, containment and the removal and disposal of all debris from the site. Payment shall also include all labour, material and equipment for the supply, installation and maintenance of temporary let downs, and disposal of all asphalt material from the Site.

413.03.04 Removal of Concrete for Partial and Full Depth Repairs – The Contractor, in the presence of the Ministry Representative, shall carry out a detailed visual inspection of the deck surface for patch concrete. A chain drag method shall be used to examine the entire deck surface for evidence of additional delaminated, spalled, and/or deteriorated concrete. These areas of unsound concrete and patch concrete shall be removed to sound concrete or removed full depth when considered necessary by the Ministry Representative.

Removal and repair of a specific area will not be completed until the Contractor and the Ministry Representative have inspected the affected areas requiring repair, and have agreed on the extent of repair required.

The Contractor shall supply the necessary labour and supplies to mark out the repair areas.

(a) Repair Procedures – Partial and full depth concrete deck repairs shall proceed as follows:

(i) The Contractor shall keep repair boundaries square or rectangular and avoid abrupt changes in width of a given repair area. The edge of the repair area shall be vertical. No debris shall be permitted to fall from the bridge. Saw cut the perimeter of the repair areas to a minimum depth of 25 mm to prevent feather edges. Do not cut the existing reinforcing steel; Adjust saw-cut depths as required to avoid cutting reinforcing steel.

(ii) Remove deteriorated, delaminated and patch concrete to sound concrete. Maximum jack-hammer weight shall be 14 kg. Care shall be taken to avoid damage to the reinforcement and remaining concrete. Jackhammers are to be operated at angles less than 45° as measured from the surface of the deck to the hand tool. The Contractor shall use wedge type chisel bits. Moyles, bush hammers, and round point chisels are not permitted. Adhering concrete shall be removed from the reinforcement using hammers not larger than 7 kg.

(iii) Remove all concrete to allow a 25 mm space around all exposed reinforcing bar within partial depth repairs. Expose all corroded reinforcement at edges of partial and full depth repair area.

(iv) Full depth repairs will only apply when the soffit concrete is removed. For full depth repair areas, the Contractor shall install all necessary formwork prior to cleaning of reinforcing and concrete surfaces. The finished soffit repair concrete shall be flush with the surrounding soffit concrete. Areas for full depth repair exceeding 1 m^2 shall be patched prior to placement of the overlay. Before placement of the concrete patch, the surface of the adjoining concrete should be saturated with water for 30 minutes minimum and coated with a bonding agent immediately ahead of the fresh concrete.

(v) Mildly corroded reinforcement shall be blast cleaned to commercial blast finish SSPC-SP6. Severely corroded and damaged reinforcement shall be replaced. Mechanical couplers acceptable to the Ministry Representative or laps in accordance with CSA-S6, "Canadian Highway Bridge Design Code" Clause 8.15.10.1 shall be used to attach new bars to the existing bars. Mechanical couplers shall develop at least 125% of the specified yield strength of the bar and shall be in accordance with CSA-S6.

(vi) Prior to the placement of the partial and full depth repair area infill concrete the surfaces of the sub deck within the repair area shall be abrasive blast or high-pressure water blast (minimum 35 MPa / 5000 psi) cleaned to remove all bruised and fractured concrete and foreign materials such as dirt, dust, laitance, sand, grease, oil, concrete slurry etc. to the satisfaction of the Ministry Representative.

(vii) The Contractor shall collect and remove all dirt, dust, and blasting sand from the repair areas of the sub deck, by vacuum or similar method, and dispose of the waste materials away from the Site. Disposal shall be in accordance with all applicable environmental regulations.

(b) Payment – Payment for removal of Concrete for Partial Depth Concrete Repair of the sub deck will be made at the Unit Price per square metre bid. Payment shall be for all labour, materials and equipment to break out the concrete, including; inspections, sounding, cleaning of mildly corroded reinforcing, abrasive blast cleaning of the existing concrete

surfaces within the repair areas, saw cutting, and containment and disposal of all debris.

Payment for Removal of Concrete for Full Depth Concrete Repair of the sub deck will be made at the Unit Price per square metre bid. Payment shall be for all labour, materials and equipment to break out the concrete, including inspections, sounding, cleaning of mildly corroded reinforcing, abrasive blast cleaning of the existing concrete surfaces within the repair areas, saw cutting, and containment and disposal of all debris. Payment shall also include the supply, installation and removal of all required formwork.

413.03.05 Reinforcing Steel – Reinforcing steel shall be supplied and installed in accordance with SS 412. Welding of reinforcing bars shall not be permitted.

Severely corroded reinforcing steel in the existing deck, more than 25% section loss as determined by the Ministry Representative, shall be removed and replacement bars spliced to the existing reinforcement using mechanical couplers acceptable to the Ministry or lapped splices. Mechanical couplers and lap length shall be in accordance with CSA S6.

Reinforcing steel damaged by the Contractor's operations shall be replaced at the Contractor's expense.

(a) Payment – Payment for replacement of severely corroded reinforcing steel will be made at the Unit Price per kilogram shown for increases and reductions in SS 412.91 and will be paid out of the Provisional Sum Item for Site Modifications. Payment shall be for quality control, supply, cutting, bending, shipping, placing and securing of reinforcing steel.

Payment for mechanical couplers will be made on a Force Account Basis under the Provisional Sum Item for Site Modifications.

413.03.06 Removal and Replacement of Existing Deck Joints

(a) Expansion joint system – The Contractor is responsible for demolition and removal of existing concrete and expansion joints.

The steel portions of the replacement deck joint shall be supplied and fabricated in accordance with SS 422.

The steel portions of the deck joint shall be galvanized and the joint armoring shall be installed 5 mm below the finished concrete nosing surface. Galvanized surfaces damaged by welding shall be touched up with two coats of an approved zinc rich coating.

Each joint seal shall be supplied in a single length, without splices. Before the joint seal is installed, the joint shall be thoroughly cleaned with a wire brush and all moisture removed from the joint. The seal shall be installed in accordance with the Manufacturer's recommendations and demonstrated to be watertight.

(i) Payment – Payment for deck joints will be made at the Price bid for each joint. Payment shall be for all Work as shown on the Drawings and Special Provisions. Payment shall include but not be limited to demolition and removal of the existing deck and curb concrete, removal of the existing joints seal, preparation of shop drawings; the supply, fabrication, galvanizing, and installation of the joint armouring, anchors, forming, reinforcing and supply and installation of joint seal. Payment shall also include the removal of the joint curb seal section and replacement with a joint sealant within the curbs section.

(b) Preparation of Filled Joints – The Contractor shall cut a transverse line, directly above the filled joint location, during casting of the deck overlay concrete using an edge cutter attached to a bull float. The line shall be neat and straight.

The Contractor shall further saw cut the filled joint along the transverse line within 48 hours of casting the deck. The saw cut will be achieved using a 4 mm wide diamond blade to a depth of 30 mm below the top of the deck.

The Contractor shall fill the saw-cut with, a one-component, self-leveling, polyurethane sealant (such as Sikaflex 1C SL) or an alternate permanent filler, impervious to water and resistant to ultraviolet radiation, acceptable to the Ministry Representative. Sealants shall be installed in accordance with the manufacturers' recommendations. The saw cut shall be clean and dry before placing the sealer.

(i) Payment – Payment for preparation of filled joints will be made at the price bid for each joint. Payment shall be for all labour, materials and equipment required to produce the filled joints

413.03.07 Preparation of the Sub Deck – The Contractor shall high pressure (not to exceed 110 MPa/15000 psi using a rotating head) water blast the surface of the entire concrete subdeck including both parapet and curb faces 50 mm up from original deck elevation and abutment ballast walls, no earlier than 2 days before placing the overlay concrete.

The water blasting equipment shall effectively remove laitance, loose materials, bruised concrete, dust, slurry, oil, or other contaminants (as applicable) that are detrimental to the concrete overlay bond and shall leave behind a sound concrete surface, thoroughly cleaned and roughened to partially expose the coarse aggregate. The Contractor shall provide full containment and disposal of all debris, contaminants, etc., and shall comply with all applicable environmental, Workers Compensation Act, and Occupational Health and Safety Regulation requirements. The Contractor shall provide adequate guards to contain over spray and flying debris during the scarification, water blasting, deck washing and chipping process. The

Contractor shall take all means necessary to reduce and contain the amount of dust produced throughout the entire project.

For the period between water blasting and placement of overlay concrete, the sub-deck shall be protected from contaminants.

Water blasting equipment shall travel downhill, keeping all waste material ahead of the equipment and preventing the waste material from rehydrating on the previously cleaned surfaces.

(a) Payment – Payment for high pressure water blasting the entire deck surface, including both parapet faces and abutment ballast walls, to clean and roughen the sound concrete surface, will be made at the Lump Sum price bid.

Payment shall be for all labour, materials and equipment required to produce the specified surface roughness. Payment shall also be for the containment and treatment of the contaminated water and the removal and disposal of any debris from the Site.

413.03.08 Concrete Overlay Placement – All concrete works will be governed by the requirements of SS 211 "Portland Cement Concrete" and SS 413.02, except as otherwise specified in this Section.

The Contractor shall be responsible for the design of all concrete mixes. The Contractor shall be responsible for quality control of each component of the concreting operation, including aggregate and component quality, batching, mixing, transporting, placing, consolidating, finishing, curing and testing.

The Contractor shall submit a detailed Overlay Placement and Curing plan of equipment and manpower to be used for placement and curing of the concrete overlay a minimum of 14 days prior to the scheduled placement dates for the acceptance of the Ministry Representative. The plan shall also contain details such as: water source, the distribution system, the collection and run-off control system, person(s) responsible for quality control, and staff to maintain the system. All review concerns raised by the Ministry Representative shall be addressed to the satisfaction of the Ministry Representative.

The Contractor will undertake a test placement of the proposed deck mix design as indicated by the requirements of SS 211.03.05.

(a) Materials & Equipment – The only acceptable type of overlay is silica fume modified concrete**.**

For silica fume modified concrete overlays, acceptable finishing machines are Terex Bid-Well models 2450, 3600 or 4800, Gomaco models C-A450, C-750, Allen Models 4836B, 6036 B 6048 B or alternate acceptable to the Ministry Representative.

Concrete shall be produced either by a qualified concrete batch plant or at the bridge site using pre-bagged concrete mix and mixer trucks. All equipment shall be provided in good working condition.

Finishing machines shall be capable of forward and reverse motion under positive control and with provision for raising screeds to clear previously screeded surfaces while traveling in reverse.

The Contractor shall provide steel rail mounted mechanical deck concrete finishing equipment of adequate size and design to permit the complete placement and finishing of a single lane section from abutment to abutment with the prescribed surface finish, without forming any cold joints.

The Contractor is responsible for properly setting the screed rails to ensure longitudinal and transverse drainage from the deck without ponded areas or "bird baths".

Sufficient screed guide rails will be set out for the full length of the anticipated section to be cast plus 6 m length at each end for run in and run out of the deck machine, adjusted for height and accepted by the Ministry Representative prior to the pour as follows:

(i) Screed guide rails, upon which the finished machine will be placed outside the area to be concreted, will be horizontally and vertically stable. Rails shall be installed with fully adjustable supports at 350 mm maximum spacing

(ii) The finishing machine and guide rails will be adjusted so that the height of the screed above the existing subdeck at each point meets the Ministry Representative's requirements. To confirm the adjustment of the machine and guide rails, the screed will be "dry-run" and clearance measurements taken at 3 m intervals and provided to the Ministry Representative for acceptance. The minimum overlay thickness shall be 50 mm. Resetting of the machine and/or guide rails will be done as necessary to obtain an acceptable "dry-run". Adjustments to the machine or the rails will not be permitted after an acceptable "dry-run" is complete.

Work bridges shall be provided to facilitate bull floating, edge finishing with hand tools, correction of defects in machine finished concrete and tining and fog curing.

Rigid forms made of wood or steel shall be installed at longitudinal bulkheads in a manner that will maintain the desired shape during concrete placement, consolidation and finishing. The longitudinal bulkhead will be installed at the centerline or lane lines of the bridge deck. The top of the longitudinal bulkhead will be set at the finished deck elevation. Any

honeycombing or poorly consolidated concrete after stripping the longitudinal bulkhead will be cut back into the new overlay a minimum of 150 mm.

The overlay shall be placed in strips with longitudinal joints near lane markings. Each strip shall be placed continuously between joints; adjacent strips shall not be cast for 24 hours.

(b) Bonding Agent – The Contractor shall apply a pre-approved cementitious bonding slurry to the prepared sub-deck and repair concrete. The prepared concrete sub deck shall be in a saturated surface dry (SSD) state prior to slurry placement. The Contractor shall supply the necessary equipment such as water hoses and vacuum or oil free compressed air.

The cementitious bonding slurry is to consist of type GU or GUL portland cement mixed with a water/cement ratio not exceeding 0.38 by mass. The slurry must not exceed 3 mm in depth and be scrubbed into the substrate surface with coarse brooms; in particular, no pooling will be permitted. The bonding slurry is to be applied immediately ahead of the overlay concrete placement and in a timely fashion so as not allow drying or setting up of the slurry prior to the placement of overlay concrete. Any areas to which the slurry has been applied, that in the opinion of the Ministry Representative exhibit drying or setting shall be cleaned of the bonding slurry and re-coated prior to the placement of deck overlay concrete.

The vertical faces at the edge of the scarified sub-deck surface; at construction joints; and at concrete/deck-joint interfaces for old bridge decks and the vertical faces at sub-deck and curb or parapet interface for new decks shall be coated with an epoxy-bonding agent conforming to ASTM C881, Type V, and Grade 2. The Class of bonding agent shall be appropriate for the temperature at the time of application. The bonding agent shall be applied in accordance with the manufacturer's instructions.

(c) Interim Partial Strength – Overlay shall attain a minimum strength of at least 30 MPa before removal of the burlap and opening to traffic. All other requirements of seven day curing shall be adhered to. The interim strength tests shall be determined using field cured test cylinders cured in accordance with Clause 9.3.2 of CSA A23.2-3C.

(d) Payment – Payment for placement of the concrete overlay and concrete for partial and full depth repairs will be made at the Unit Price per cubic metre bid. Payment shall be for all labour, materials and equipment for the final preparation and cleaning of the existing prepared concrete sub deck, supply and application of the cementitious bonding agent and the epoxy bonding agent, the supply of cement, silica fume, superplasticizer, all other constituent concrete materials and the supply and placing and curing of the concrete and containment and disposal of all debris.

SECTION 415

MANUFACTURE AND ERECTION OF PRECAST AND PRECAST PRESTRESSED CONCRETE MEMBERS

DESCRIPTION

415.01 Scope – This Section covers the manufacture and erection of precast and precast prestressed concrete members.

415.02 General

415.02.01 415.02.01 General Requirements – The manufacture and erection of precast and precast prestressed concrete members shall conform to the requirements of these specifications and details as shown on the Drawings and in the Special Provisions.

415.02.02 Plant Certification – Precast concrete elements shall be manufactured in plants certified to the current CSA A23.4 requirements for the product groups corresponding to the element types being fabricated. Certification shall be by the Canadian Precast Concrete Quality Assurance (CPCQA) Certification Program or by a certification organization accredited by the Standards Council of Canada (i.e. Canadian Standards Association (CSA)). Certification shall be in effect prior to the beginning of Work, and maintained throughout the period of manufacture.

415.02.03 Materials and Construction – Materials and construction shall conform to the current CSA A23.4 and the current PCI "Manual for Quality Control for Plants and Production of Structural Precast Concrete Products". Where there is a discrepancy between the requirements of these two standards, the more stringent shall apply. In cases where it may be unclear as to which standard is more stringent, the Ministry Representative in consultation with the Ministry Chief Bridge Engineer will have the final say.

415.03 Prefabrication Meeting – The Ministry will at its discretion, convene a prefabrication meeting with the Contractor to review issues such as, but not limited to, lines of communication, fabricator and sub-fabricator scope of work, location of all work, procedures on quality control, plant certification, concrete source and mix designs, use of Hydration Stabilization Admixtures (if applicable), mill certificates, alternate details and procedures , and any other specific requirements of this specification as it relates to the Work.

415.04 Working Drawings – Working drawings shall consist of the following:

- Shop drawings,
- Transportation details, and
- Erection drawings

Transportation details and erection drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of the Province of British Columbia (APEGBC).

Shop drawings shall be prepared and sealed by a professional engineer registered with APEGBC when the Contractor is responsible for the design of items that are detailed on the shop drawings.

Working drawings shall be in the same system of units as the design drawings.

Lettering for notes and dimensions shall be at least 2.5 mm and 4 mm for headings. Drawings shall be legible when printed on 11" x 17" sheets.

415.04.01 Shop Drawings – Shop drawings shall show all information and details needed for the fabrication of the members including, but not limited to, such items as member shapes and dimensions, mark numbers and general arrangement of member locations, mass, prestressed and non-prestressed reinforcement, embedments, openings, block outs, chamfers, recesses, finishes, concrete mix design, stressing details, special tolerances, special handling instructions, lifting details, and lifting locations.

A copy of the shop drawings shall be available at all times at the location where the components shown on the drawings are being fabricated. Changes to the precast from what is shown on the reviewed shop drawings, or repairs made during fabrication and/or construction, shall be indicated by the Contractor on a marked-up set of shop drawings and submitted in digital format to the Ministry Representative at the completion of the Work.

415.04.02 Transportation Details – Transportation details shall include such items as:

(a) Description of hauling and handling equipment,

(b) Weight of members,

(c) Length and height of loads,

(d) Location and method of member support, and engineering calculations for situations where members are to be supported during transportation further from their ends than outlined in SS 415.48,

(e) Details for handling, storing, and loading of members.

415.04.03 Erection Drawings – Erection drawings shall show in detail the method of erection including, but not limited to, the following:

(a) Erection procedures

(b) Procedures for off-loading of members upon delivery

(c) Details for temporary storage and support of members on site prior to erection

(d) Equipment to be used

(e) Layout or general arrangement drawing showing the layout of the members, equipment positioning, and access roads

(f) Crane make, model, and capacity charts, boom length(s), crane placement, and access for transporting of members to crane(s)

(g) Radii and loads for crane lifts

(h) Rigging details

(i) Mass of members, rigging and special installation equipment

(j) Details for installation and removal of all falsework, temporary supports, temporary bearings, bracing, guys, dead-men, and lifting devices

(k) Attachments to the bridge members and bridge structure for temporary support and special launching equipment

(l) Detailed description of sequence of operations

(m) Details for special installation equipment such as a launching truss, launching nose, head frames, spreader beams and rollers

(n) Details for installation of members onto the permanent bearings

(o) Traffic control plan for roadway and rail traffic

(p) Details for protection of existing utilities affected by the erection procedures

(q) Layout and detail of fall protection and their sequence of installation.

The Contractor shall be responsible for the lateral stability of members and shall design and provide bracing as necessary until completion of the Work.

The erection drawings shall be complete in detail for all anticipated phases and conditions during erection and during the temporary support of members. The Contractor shall submit calculations upon request, to the Ministry Representative that demonstrate that allowable stresses are not exceeded in members, falsework, temporary bracing and temporary supports and that member capacities and final geometry will be correct. The calculations shall be sealed by the professional engineer that sealed the erection drawings.

Falsework, temporary supports and temporary bracing shall meet the requirements of CSA S269.1, "Falsework and formwork" and shall also meet all the requirements for falsework given in Clauses 20.17 to 20.26 inclusive of the WCB Occupational Health and Safety Regulation.

A professional engineer registered with APEGBC shall be responsible for any field designs and any changes made to the erection procedures. Field designs and changes to the erection procedures must be documented and sealed by the responsible professional engineer and must be available at the Site prior to the affected erection work being carried out.

Immediately before placement of loading on falsework, the Contractor must ensure that the falsework is inspected and a sealed engineering certificate is issued by a professional engineer registered with APEGBC which:

- Indicates the specific areas inspected, and
- Certifies that the falsework has been erected in accordance with the latest approved erection drawings and supplementary instructions.

Prestressed concrete stringers shall not be set on their final bearings, nor shall the bridge deck be cast, until 30 days after the casting of the stringers. If it is necessary to erect the stringers earlier, they shall be placed on temporary supports. Prestressed concrete twin cell box stringers up to a maximum length of 20 metres may be shipped and erected after 10 days if the specified 28 day compressive strength has been attained.

415.04.04 Submittals – The Contractor shall submit working drawings in digital format to the Ministry Representative at least 14 days prior to commencement of fabrication. These working drawings shall have been reviewed and approved by the Contractor. By this review and approval, the Contractor represents that it has determined and verified all field measurements, field construction criteria, materials, and similar data, and that it has checked and coordinated each working drawing with the requirements of the work and the contract documents. The Contractor shall indicate its review and approval by including on each drawing the date and signature of a person designated by the Contractor as being responsible for the Work. Working drawings shall be accompanied by a transmittal listing each of the drawings submitted. At the time of submission, the Contractor must notify the Ministry Representative in writing of any deviations in the shop drawings from the requirements of the contract documents. Any Work done or materials ordered prior to the review of the working drawings shall be at the Contractor's risk. The Ministry Representative will review the drawings for general compliance with the contract requirements.

If modifications to the drawings are required, the Ministry Representative will return one set of drawings, marked up, to the Contractor. The Contractor shall re-submit one complete set of revised drawings in digital format to the Ministry Representative. Any drawing that has changed from the version originally submitted shall be identified as such on a transmittal accompanying the revised drawing set.

If no exceptions are taken to the drawings, the Ministry Representative will return one set of reviewed drawings to the Contractor. Shop drawings will not be reviewed without

the transportation and erection drawings applicable to the members in question.

Erection will not be allowed to proceed without the Ministry Representative's review of the method proposed.

Review of working drawings shall not relieve the Contractor of any responsibility for dimensions or detail or for carrying out the work in full accordance with the drawings and specifications.

If so agreed to in advance by the Ministry Representative, working drawings may be submitted in paper format. The reference to working drawing submittal copies shall be increased to four paper copies in this case.

At least 14 days before fabrication is to commence, or as otherwise requested by the Ministry, the fabricator shall submit a schedule of fabrication to the Ministry Representative in the form of a Gantt Chart. At the discretion of the Ministry, the schedule shall be updated on no less than a monthly basis. The schedule shall be provided to the Fabricator's Quality Control and to the Ministry Representative for reference and planning of inspections and progress reporting. The Contractor shall report any interim variations to the schedule.

415.04.05 Working Drawing Format – The Contractor shall transmit working drawings through attachments to e-mail. Unless otherwise agreed to by the Ministry Representative, electronic attachments to an email must total no more than 20 MB and must be submitted unzipped. Drawing files shall be submitted in PDF format to print out on 11" x 17" size pages. PDF sets shall be created by "distilling" CAD sheets rather than by scanning paper plan sets. Unless otherwise agreed to by the Ministry Representative, electronic attachments greater than 20 MB in size shall be sent in two parts by separate emails, denoting "1 of 2" and "2 of 2" in the subject lines after other required subject-line information.

If agreed to by the Ministry Representative, the Contractor may employ a document and data management service such as SharePoint® to transmit working drawings. If this process is used, the limitation on drawing file size is waived. The Contractor shall be responsible for setting up the appropriate folders within the document and data management software and for providing access to these folders to the fabricator, Ministry Representative and the design engineer. Email notifications shall be sent to all parties whenever updates are made.

The resolution of drawings shall be such that the finest detail must be legible at full scale on a monitor without zooming in (1-in. width on an 11 x 17-in. sheet is 1 in. on the monitor).

Drawings shall be black images on a white background.

All PDF sheets within a single file shall be assembled ensuring that all sheets are rotated to a "ready to read" orientation within the PDF file set. Generally, 11" x 17" plan sheets in landscape, 8½" x 11" note sheets in portrait, so that the majority of text is vertical. PDF sheets shall be ready to print out on appropriately sized paper sheets with no additional formatting required by the viewer.

Drawings of large or complicated pieces, where it is not practical to show all details on an 11" x 17" sheet, may be submitted on full size paper drawings, in which case the Contractor shall submit four copies of the drawing set. When revisions to full size paper drawings are required, the Contractor shall supply four complete sets to the Ministry Representative.

415.05 Alternative Details – All details shall, in general, conform to those shown on the Drawings. Any proposed variation shall be submitted in accordance with the provisions of SS 125 – Value Engineering Proposals. The submission of drawings showing alternative details shall be prepared and sealed by a professional engineer registered with APEGBC.

Alternative details must meet the requirements of the Contract and shall require the acceptance of the Ministry Representative prior to implementation.

Any consideration or implementation of a submitted VEP by the Ministry will also be in accordance with SS 125.

415.06 Quality Control – The Contractor shall implement a quality control program to meet the Contract requirements. The quality control plan shall be made available to the Ministry Representative for review.

415.07 Quality Assurance – The Ministry will implement a quality assurance program by auditing the Contractor's quality control program and by inspection at its discretion.

The Contractor shall notify the Ministry Representative at least 14 days before fabrication is to commence. In addition, the Contractor shall provide a minimum of 48 hours notice to the Ministry Representative that a product will be available for inspection and/or certification by the Ministry's Inspector. If the schedule is subsequently changed, the Contractor shall provide the 48 hour notice from the time that the Ministry is notified of this change. If the product is not available or is not sufficiently complete for inspection/certification as notified, at the sole discretion of the Ministry Representative, the Contractor shall be charged stand-by costs for the Ministry's Inspector. The Contractor shall allow the Ministry's representatives safe access to all parts of work and shall supply such information and assistance as is required. When the Ministry requests, the Contractor shall provide samples of any materials. Inspection by the Ministry Representative shall not relieve the Contractor from obligation to perform the work in accordance with the Contract.

Concrete tests shall be in accordance with the current CSA A23.2 and with concrete sampled as close to the point of final deposit into the form as is practicable. All testing

personnel shall be certified under CSA A283 or certified as an ACI Concrete Field Testing Technician – Grade 1 (minimum). The Contractor shall make available at the plant a sheltered, heated space for the casting and storage of test cylinders, with curing tanks and a concrete cylinder testing machine.

The Ministry may reject any items which, in its opinion, do not comply with the requirements of this specification. The Ministry in its sole discretion may back charge all inspection costs for the rejected material to the Contractor.

Each unit shall be certified as acceptable by a Ministry Inspector before it is shipped from the shop. This certification shall not relieve the Contractor of responsibility for subsequent damage or for defects which become apparent before the work is finally accepted by the Ministry. The Contractor shall provide 5 days notice to the Ministry Representative regarding intent to ship a unit or product and the product shall be made available for inspection prior to loading and shipping.

MATERIALS

415.11 Materials Supplied by the Contractor

415.11.01 Cementitious Material – The Contractor shall obtain, from the supplier, a mill test data stating the product meets the requirements of the applicable standard representing each type of cementitious material for each month period representing the delivered shipment.

(a) Hydraulic Cement – Unless otherwise specified, the Contractor shall use Type GU (General Use) or Type GUL (General Use Limestone) Cement in all concrete. GU and GUL cements shall conform to the requirements of CSA A3000. Cement shall have a C3A content greater than 4.0% (by standard Bogue calculation) or 5% by mill certificate.

The total alkali content [Na_2O equivalent, (Na_2O + (0.658 x K_2O))] of the cement shall not exceed 0.60% by mass.

(b) Silica Fume – Condensed silica fume shall conform to the requirements of CSA A3000 for Type SF silica fume.

(c) Fly Ash – All fly ash shall conform to the requirements of CSA A3000 for Type F fly ash.

(d) Slag – All slag shall conform to the requirements of CSA A3000 for Type S ground granulated blast furnace slag.

(e) Blended Hydraulic Cement – Blended Hydraulic Cement shall not be used without the acceptance of the Ministry Representative.

415.11.02 Aggregates – Fine and Coarse Aggregates shall conform to the current CSA A23.1. Nominal maximum size of coarse aggregates shall not exceed 20 mm.

415.11.03 Water – Water shall conform to CSA A23.1. The mortars and tests shall be made in accordance with CSA A23.2.

415.11.04 Reinforcing Steel – All reinforcing steel shall be in accordance with SS 412.

All reinforcing shall be of the lengths called for on the Drawings and all bends shall be made in accordance with SS 412.

All bending schedules shall be furnished by the Contractor for the Ministry's review and any material ordered before such review shall be at the Contractor's risk. The review of bending diagrams by the Ministry shall in no way relieve the Contractor of responsibility for the correctness thereof.

415.11.05 Prestressing Steel – Pre-tensioning steel shall consist of seven-wire, uncoated, low relaxation strands as called for on the Drawings and complying with the requirements of ASTM A416M.

All strands for each project shall be obtained from one mill, and preferably strands from one heat shall be used for the Work. Members shall use strands from one heat. When multiple heats are used in a project, the modulus shall be matched in all members. Each reel of strand shall be identified by a secure tag showing the size of strand, CSA or ASTM Designation number, heat number, and name or mark of the manufacturer.

The Contractor shall supply to the Ministry Representative, for each heat number, one set of the standard mill test reports. Strands shall be protected at all times to prevent corrosion and shall be dry and clean (free from scale, rust, oil, soap, grease, and other deleterious materials) immediately before the concrete is placed. No strand which has previously been tensioned or gripped by strand chucks shall be incorporated into the members.

415.11.06 Admixtures – Chemical admixtures shall conform to ASTM C494. Air-entraining admixtures shall conform to ASTM C260.

Water reducing admixtures shall be of a lignosulfonic acid or salt, or hydroxylated carboxylic acid or salt.

Type B, Retarding or Type D, Water-Reducing and Retarding (Hydration Stabilizing) Admixtures conforming to the requirements of ASTM C494 shall not be incorporated into the mix design and/or added to the concrete without the acceptance of the Ministry Representative. When accepted by the Ministry Representative, the guidelines given in SS 415.37, SS 415.40.01 and SS 415.42.01 shall apply.

415.11.07 Mix Design Submission and requirements – The Contractor shall submit a report outlining the proposed

mix design for each classification of concrete to the Ministry Representative for review and acceptance in accordance with SS 211.03. Trial mix test batches for precast members are not required provided that the sampling, testing, and inspection of concrete for qualification purposes has been completed in accordance with A23.2-24C.

415.12 Storage of Materials – All materials shall be stored in accordance with the requirements of CSA A23.1. Cement shipments shall be used in the order in which they are received.

415.13 Premixed Aggregates – Pre-mixed aggregates in which the fine and coarse aggregates are combined in definite proportions will be permitted provided that the aggregates are delivered in batches directly to the hopper of the concrete mixer. Pre-mixed aggregates shall be transported and delivered in batches, each containing the correct quantity for one batch of concrete. At no time after the fine and coarse aggregates are combined shall any batch come in contact with or intermingle with any other batch.

415.14 Forms – All exterior forms shall be of steel and shall be mortar-tight and of sufficient rigidity to prevent distortion due to incidental loadings during construction. Forms for skewed ends may be of rigid steel-reinforced plywood.

Wood forms will be permitted for the side walls of ballast wall pieces and for side walls of the deck panel pieces only. Wood formwork shall be designed, supplied and installed in accordance with CSA-S269.1. Wood formwork shall render a true smooth surface free from fins and projections. Only new plywood shall be permitted for plywood formwork.

Forms shall produce 20 mm chamfers and fillets as shown on the Drawings.

415.15 Ministry's Office – Except for "on site" precasting work, where an office is already provided under the general contract, the Contractor shall provide at the point of manufacture a suitable lock-up office for the sole use of the Ministry.

The office shall be weatherproof and reasonably soundproof and provided with ample window area and ventilation. Location of the office shall be acceptable to the Ministry Representative. The office shall measure not less than 3 m x 4 m. If the office is to be used concurrently with other inspection agencies, the minimum area required for Ministry's use shall be 3 m x 4 m. The Contractor shall install satisfactory heat and lights and provide a telephone, high-speed internet access, and a draughting table 1 m x 2.5 m in size with a plywood top, desk 0.75 m x 1 m, two chairs, drafting stool and one square metre of shelves.

Long distance calls made by Ministry personnel will be to the Ministry's account.

The Contractor shall, during the life of the work, regularly clean and properly maintain, heat and light to the office.

The office and contents shall be for the use of the Ministry Representative for the duration of the Work and may, if necessary, be used concurrently with other inspection agencies.

CONSTRUCTION

415.31 Reinforcing Steel – All reinforcing bars shall be accurately placed in the positions shown on the Drawings and firmly held during placing and setting of the concrete. Installation shall be in accordance with SS 412 and this specification. Tolerances shall be as specified by CSA A23.4.

When placed in the work the reinforcing steel shall be free from dirt, rust, loose scale, paint, oil, or other foreign materials. Bars shall be tied at all intersections except that where spacing is less than 300 mm, alternate intersections shall be tied. Form clearance shall be maintained by stays, ties, hangers or other approved supports. Before placing concrete, the placing and securing of reinforcing in the forms shall be inspected by the Ministry Representative.

All reinforcing shall be placed in the full lengths called for on the Drawings. Splicing of bars will not be permitted unless acceptable to the Ministry Representative. Splices shall be staggered. Unless otherwise shown or called for on the Drawings, splices shall be made by lapping in accordance with the requirements of SS 412. The bars shall be placed so as to maintain the minimum specified clearances between bars and the concrete surface. Welding of reinforcing steel shall be done only if called for on the Drawings.

Tack welding of reinforcing steel, if permitted, shall be in accordance with CSA W186-M.

415.32 Stressing – The dials of gauges shall be at least 150 mm in diameter; digital gauges are an acceptable alternative. Gauges shall be calibrated with the hoses and jacks with which they are to be used, in accordance with CSA A23.4 clause 28.2.3.2. The calibration must have been performed within the previous 6 months of use or 12 months maximum. Before stressing commences, certified calibration curves shall be furnished to the Ministry Representative. All strand chucks shall be cleaned and carefully inspected before each use.

At the sole discretion of the Ministry, the stressing operation may be inspected by the Ministry Representative. The Contractor shall provide a minimum 48 hours notice to the Ministry Representative prior to the stressing operation to allow for inspection. If during the work any gauges give erratic results, the tension system or tension device shall be recalibrated.

The Contractor shall schedule the work so that placement of concrete follows tensioning of strands within 24 hours. In cases of unavoidable delay, a maximum period of 80 hours will be allowed, provided that the strands do not show unacceptable corrosion (see current PCI "Manual for Quality Control for Plants and Production of Structural Precast Concrete Products"). The reason for the unavoidable delay shall be documented in the Contractor's Quality Control documentation along with documentation that strand condition is still acceptable. This information shall be made available to the Ministry Representative upon request.

Before tensioning, an initial force shall be applied to each strand to equalize stresses and eliminate slack. This force shall be between 5 and 25% of the final prestress force. After the initial tension forces have been applied, each strand shall be marked at the outer end of each chuck as a reference line from which the elongation of the strand can be measured.

The method of tensioning shall be such that the specified tension is maintained in the strands until the forces in the strands are transferred to the concrete member(s). The maximum jacking load in each strand shall be 80% of the ultimate strength of the strand.

A tensioning force shall be determined both by reading the jacking load on the calibrated gauge and by measuring the elongation of the strand. When the difference between the two methods is greater than 5%, the cause of the discrepancy shall be ascertained and corrected.

Elongations of strands shall be calculated and shown on the working drawings. Calculations for elongations shall include appropriate allowances for possible slippage or relaxation of the anchorages, friction at hold down points, temperature difference, and strand relaxation.

415.33 Forms – The design of the forms shall take into account the effect of vibration on concrete as it is placed.

All exterior forms shall be set and maintained true to the designated lines until the concrete has hardened. When forms are unsatisfactory in any way, either before or during the placing of concrete, the Ministry Representative may order the work stopped until the defects have been corrected.

The shape, strength, rigidity, mortar-tightness and surface smoothness of re-used forms shall be maintained at all times. Any warped or bulged formwork must be realigned before being re-used. Forms which are unsatisfactory in any respect shall not be reused.

Void forms shall be of an approved product and shall be anchored firmly by a method acceptable to the Ministry Representative to prevent displacement during the placing of the concrete. Any member with a void out of position in excess of the specified tolerance will be rejected. Exterior forms shall be treated in moderation with form oil before placing the concrete. Any material which will adhere to or discolour the concrete shall not be used.

Side forms and any attachments which would obstruct strain of the members shall be removed before transfer of stress.

415.34 Admixtures – The Contractor shall add sufficient air-entraining agent to ensure an air-void system in accordance with CSA A23.1 or as called for on the Drawings, and as determined by an air-meter used in accordance with CSA A23.2.

The use of other admixtures, such as water-reducing agents or superplasticizer, shall be subject to the acceptance of the Ministry Representative.

Type B, Retarding or Type D, Water-Reducing and Retarding (Hydration Stabilizing) Admixtures shall not be incorporated into the mix design and/or added to the concrete without the acceptance of the Ministry Representative. When accepted by the Ministry Representative, the guidelines given in SS 415.37, SS 415.40.01 and SS 415.42.01 shall apply.

415.35 Measurement of Materials – Measurement of materials shall conform to CSA A23.1. The Contractor shall produce a certificate or report of inspection from a weights and measures approved, independent testing authority, and which is not more than 6 months old. The certificate or report will be accepted as proof of accuracy for the batching equipment. Cement in standard packages need not be weighed.

The surface moisture shall be determined in accordance with CSA A23.2 and the batch weights of aggregates and water adjusted accordingly.

415.36 Mix Requirements – Minimum compressive strength at time of transfer of stress to the concrete and at 28 days as determined by CSA A23.2 shall be as called for on the Drawings. In any case, the force in the prestressing tendons shall not be transferred to the concrete until the compressive strength of the concrete is at least 25 MPa for pretensioned components and 20 MPa for post-tensioned components.

Slump measurement shall be determined by CSA A23.2. The mix design shall specify the slump and the allowable variation.

To minimize shrinkage and creep, the cement/aggregate ratio shall be kept as low as is consistent with strength requirements.

415.36.01 Self Consolidating Concrete (SCC) – SCC shall meet the requirements of CSA A23.1.

For the use of SCC, the Contractor shall submit the proposed mix design and detailed documentation for ministry review, a minimum of 28 days in advance of

fabrication, which shall include as a minimum the following:

(a) Details in accordance with CSA A23.2-24C documenting the proven history of successful performance for the required strength, durability, and other performance requirements of the proposed mix design in the fabrication of products similar to those proposed for the use of SCC in the Work; or

(b) A full-scale test shall be used to verify the self-consolidating characteristics for placement and for the hardened concrete properties of the mix design for the proposed application. Documentation in accordance with CSA A23.2-24C shall be submitted demonstrating that the proposed mix design will achieve the required strength, durability, and performance requirements.

The Ministry Representative may also request additional documentation be submitted to assist in the ministry review. If, in the sole discretion of the Ministry Representative, the submitted documentation does not demonstrate acceptable performance of the SCC, then SCC shall not be used.

415.36.02 Flowing Concrete – The specified slump shall not be greater than 190 mm. When the specified slump is greater than 180 mm, the allowable variation shall be ±40 mm. Measured slump shall not exceed 230 mm.

The following requirements shall apply:

(a) Concrete vibration procedures specific to flowing concrete shall be developed to ensure proper consolidation without segregation and shall be submitted for review upon request.

(b) Personnel that have experience with the vibration procedure shall perform the Work.

(c) Qualification and production test data demonstrating a Visual Stability Index (VSI) result of 0 or 1 determined in accordance with CSA A23.2-19C.

(d) Qualification test data for the intended concrete mix, no more than 12 months old, demonstrating a Hardened Visual Stability Index (HVSI) of 0 determined in accordance with AASHTO R81.

Concrete members shall be subject to inspection for segregation after stripping. This shall be done by providing an exposed aggregate surface finish for the full depth of a portion of the member. The location of the exposed aggregate surface finish shall be chosen in consultation with the Ministry Representative and shall be in an area that will not be subject to general view by the public when the member is in place. The exposed aggregate surface shall be repaired after inspection using a thixotropic epoxy repair product acceptable to the Ministry Representative.

The exposed aggregate surface inspected for segregation shall meet the Hardened Visual Stability Index (HVSI) of 0 or 1 determined in accordance with AASHTO R81.

415.37 Batching and Mixing – The batching plant and mixing of concrete shall conform to CSA A23.1.

Mixing shall begin within 15 minutes after the cement has been added either to the water or aggregate.

Within 30 minutes after the introduction of the mixing water to the cement and aggregate, or the cement to the aggregate, the concrete shall be placed in its final position in the forms. In hot weather or under other conditions contributing to quick stiffening of the concrete, the maximum allowable time may be reduced by the Ministry Representative.

Time of placement extensions will be considered on a case-by-case basis and may require the use of Hydration Stabilizing Admixtures (HSAs) conforming to the requirements of ASTM C494 Type B, Retarding or Type D, Water-Reducing and Retarding Admixtures.

When accepted by the Ministry Representative, the use of set retarders or HSA's shall be as follows:

(a) HSA modified concrete when used shall be fully discharged and placed in its final position within 90 minutes after the introduction of water to the cement and aggregate.

(b) All concrete in a given precast element shall be placed in its final position before 90 minutes.

(c) When HSAs are used, these time extensions are subject to preconstruction trials being conducted by the Contractor, to establish the appropriate HSA dosage to provide suitable extended slump life of concrete, without increasing the water/cementitious ratio of the concrete above that which would be required if HSAs were not used.

(d) The use of HSAs shall in no instance modify the maximum concrete temperature required at time of placement.

(e) A one time only addition of HSA will be allowed. This will be during initial batching of the concrete or immediately at completion of batching as recommended by the admixture manufacturer. Addition of HSA at any other time will be cause for rejection of the concrete.

Concrete mixing and transporting equipment shall be capable of placing concrete in the forms at the rate of 20 cubic metres per hour. Batch delivery shall be uniform and at a maximum interval of 10 minutes.

The maximum size of each batch shall not exceed the maximum rated capacity of the mixer as stated by the manufacturer of the mixer.

Concrete when placed in the form shall have a temperature not less than 10°C nor greater than 30°C, except for High-